Hand and Wrist Movement Control of Myoelectric Prosthesis Based on Synergy

Jiaxin Ma, Nitish V. Thakor, *Fellow, IEEE*, and Fumitoshi Matsuno, *Member, IEEE*

***Abstract*—This study proposes a method to control a prosthetic hand by EMG signals based on muscle synergies. The muscle synergy model suggests a framework to transform commands of the central nervous system to a set of complex muscular movements. Using this method, we have tried to realize the proportional control of multiple degrees of freedom (DOF). This study focuses on controlling four kinds of hand/wrist movements of the prosthesis: open, close, pronate, and supinate. The nonnegative matrix factorization (NMF) algorithm is used to map muscle activities into these four movements through the calculation of a muscle synergy matrix. An EMG feature selection process along with a control scheme has been added, which smooths the output thereby stabilizing the movements. Ten healthy subjects performed an online experiment comprised of two tests: 1) proportional control on single DOF, and 2) simultaneous control of multiple DOFs. The results indicate that fluid hand/wrist movements could be estimated from EMG. The average R^2 values achieved by all subjects for the single-DOF test and the multiple-DOF test are 0.97 and 0.93, respectively.**

***Index Terms*—Degrees of freedom (DOF), electromyogram (EMG), nonnegative matrix factorization (NMF), prosthesis, synergy.**

I. Introduction

PROSTHETIC limbs generally available to amputees include simple grasping mechanisms. A standard one-to-one mapping control paradigm typically includes two myoelectric sites to detect electromyogram (EMG) signals from flexor and extensor muscles on the forearm. Users can only control a single degrees of freedom (DOF) of the prosthesis with the activities of these muscles. Multiple-DOF control (switching between different DOFs) needs to be realized through the incorporation of some additional commands (e.g., cocontraction). Modern prosthetic hands have multiple fingers which offer higher DOF [1]. However, EMG-based control algorithms do not have comparable control capability [2]. Higher DOFs increase the functional complexity of the prosthesis and, hence, decoding subjects' intent becomes difficult.

Pattern recognition (machine learning) methods can enhance prosthesis control. In myoelectric prosthesis related studies, various pattern recognition algorithms have been used. Artificial neural network [3], support vector machine [4], Bayesian network [5], the Gaussian mixture model classifier [6], and hidden Markov models [7] are some widely used algorithms. Decoding has also been performed for multiple DOF control cases, such as individual finger movement [8]. Through training, a pattern recognition method can "learn" various states such as "open hand" or "close hand" and recognize them perfectly. However, a prosthesis does not have limited discrete outputs. Between "open hand" and "close hand," there is a series of transition states. Proportional control of prostheses cares about how to recognize these transition states; however, training all of them is time consuming and unrealistic. Moreover, for a multiple-DOF prosthesis, even when each DOF is well trained, it does not mean that the user can simultaneously control two DOFs. For example, usually a classifier recognizes single states "open" and "pronate," but cannot respond to the combination state "open and pronate" correctly. While regression methods are more appropriate for solving these problems due to their continuous nature [9], [10], they still require supervised training. The training data must involve all of the transition states, and all of them must be correctly labeled. Such training processes are also complicated and time consuming. As a result, a method which can recognize untrained transition states or combination states, with less training time, is preferred.

Neuroscience can support building models for natural movements, where a typical representation is the muscle synergy. Unlike normal pattern recognition methods which are based on statistical rules, this theory offers an understanding of the underlying processes of limb movements; therefore, it can potentially recognize untrained states. According to the theory of muscle synergies, when a subject's central nervous system (CNS) controls the body movements, it does not send a single command to multiple muscles individually. Instead, it activates a small set of coordinated movements or a cascade of movement primitives called synergies. Each synergy is responsible for activating a larger set of muscles. The theory has been tested on frogs, mammals, primates, and humans [11]–[14]. An example of a marionette can also illustrate the basic idea: only several strings can control the complex movements of the puppet because the manipulator's hand (equivalent to the CNS) sends out higher order commands to activate the strings (synergies) and achieves control of multiple joints of the puppet (human limbs). Synergies and muscles possess many-to-many mappings, and the number of synergies needed is typically far less than the number of muscles. This approach of modeling limb motion can also be regarded as a dimension reduction mechanism from muscles to synergies. If the synergies can be calculated for an amputee, then it would be possible to represent the extent of a single movement, as well as the simultaneous activation of multiple movements.

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J. Ma and F. Matsuno are with the Department of Mechanical Engineering and Science, School of Engineering, Kyoto University, Kyoto 6158530, Japan (e-mail: ma.jiaxin.62c@st.kyoto-u.ac.jp; matsuno@me.kyoto-u.ac.jp).

N. V. Thakor is with the Department of Biomedical Engineering, School of Medicine, The Johns Hopkins University, Baltimore, MD 21205 USA (e-mail: thakorjhu@gmail.com).

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Currently, typical algorithms to calculate muscle synergies are dimension reduction algorithms such as principal component analysis (PCA) [15], independent component analysis (ICA) [16], and nonnegative matrix factorization (NMF) [17]. These three algorithms decompose a larger matrix into low dimension ones while applying certain restrictions, namely orthogonality, statistical independence, and nonnegativity, respectively. Among them, NMF is considered close to the natural representation of synergies and outperforms PCA and rivals ICA [18]. In this paper, we decoded four hand/wrist movements which are open, close, pronate, and supinate from surface EMG by using the NMF algorithm. Furthermore, we designed an original control scheme to process the noisy output of NMF, and convert it to an eligible control signal.

Ajiboye and Weir [19] previously used the NMF algorithm to classify postures. They used 11 postures as training data to predict 33 postures. They have validated the usefulness of the NMF in prediction, but their approach was discrete classification and did not show the strength of a synergy-based method. Jiang *et al.* [20] presented an offline decoding result of four-direction wrist movements (flexion/extension and radial/ulnar deviation) plus pronation/supination, by using a semisupervised NMF algorithm. They also examined the influence of muscle crosstalk and force transfer by comparing several methods. Choi and Kim [21] proposed a similar approach for four-direction wrist movements with a complete supervised NMF algorithm and carried out an online experiment with visual feedback. Jiang *et al.* validated that the NMF algorithm can effectively decode the four-direction wrist movements, but a significant performance loss was reported when dealing with pronate/supinate movement [20]. Furthermore, they did not experiment with open and closed hands. Common prosthetic applications require opening and closing of the hand, as well as wrist pronation and supination. Therefore, this study focuses on opening and closing of the hand and the pronation/supination of the wrist. Compared with four-direction wrist movements, these movements involve more muscles. If more muscles located closely are activated, the crosstalk that contaminates the surface EMG signals will be much more significant and the movements are expected to be more difficult to decode [20]. To improve the performance, an offline experiment for feature selection was first carried out to test which features are best for calculating muscle synergies.

Another objective of this study is to design a control algorithm for the prosthetic hand. Although, the synergy-based paradigm offers continuous output, this output is too variable to be directly interpreted as a control signal of the prosthesis, and thus needs postprocessing. Typically used filters are incapable of suppressing this fluctuation satisfactorily. Since the control of a prosthesis is a delicate process and users' experiences can be hampered by undesired output, it is important to generate smooth control signals. Unfortunately, few previous studies include this variety of suggested postprocessing. Their results are generally continuous but fluctuating, or are discretized approximations of the continuous output [10]. In this paper, a control scheme is designed to achieve direct control of the prosthesis. Online experiments are made to validate its reliability.

II. Methods

A. Synergy Model and NMF

When a person tries to perform a movement, his CNS continuously sends commands to activate muscle synergies. Suppose the time-varying command sent by CNS is $\mathbf{H}(t) = [h_1(t), h_2(t), \ldots, h_K(t)]^T$, where $h_i(t)$ refers to the expected activation level on the ith synergy at time t. Here, K is the total number of synergies involved in movement-related tasks. The actual value of K is unknown, but for simplification it can be assumed that one simple movement is represented by one synergy [20], [21]. This assumption becomes inappropriate when a large variety of movements are involved. In that case, a reduced set of synergies should be found [19], but this study focuses on a control paradigm; therefore, we do not go that far. Our task includes four basic movements (open/close/pronate/supinate); therefore, the number of synergies is $K = 4$, where each synergy corresponds to one of these movements.

The muscles' activation levels $\mathbf{V}(t)$ can be expressed by $\mathbf{H}(t)$ multiplying a synergy matrix $\mathbf{W}$

$$\begin{aligned}\mathbf{V}(t)_{M\times 1} &= \mathbf{W}_{M\times K} \times \mathbf{H}(t)_{K\times 1} \\ &= \begin{bmatrix} W_1 & W_2 & \cdots & W_K \end{bmatrix} \times \begin{bmatrix} h_1(t) \\ h_2(t) \\ \vdots \\ h_K(t) \end{bmatrix} \\ &= \sum_{i=1}^{K} W_i h_i(t)\end{aligned} \tag{1}$$

where $W_i = [w_{1i}, w_{2i}, \ldots, w_{Mi}]^T$. $W_i h_i(t)$ is the effect of the ith synergy to all the M muscles. For a single muscle j, its activation level $V_j(t)$ is equal to the element of the jth row of $\sum_i^K W_i h_i(t)$.

From this synergy-based model, we expect to solve the equation

$$\mathbf{V} = \mathbf{W} \times \mathbf{H} \tag{2}$$

because the values of $\mathbf{H}$ contain the details of each movement. However, $\mathbf{V}$ (unmeasurable) is neither equal to nor linearly correlated with the surface EMG signal $\mathbf{Z}$ (which can be easily measured). In this case, a common solution is to assume that $\mathbf{V}$ has an approximately linear correlation with $f(\mathbf{Z})$, which is some feature extracted from $\mathbf{Z}$ [19]–[21]. Then, (2) can be treated as

$$f(\mathbf{Z}) \approx \mathbf{W} \times \mathbf{H}. \tag{3}$$

The NMF algorithm decomposes the nonnegative matrix $f(\mathbf{Z})_{N\times O}$ into nonnegative factors $\mathbf{W}_{N\times K}$ and $\mathbf{H}_{K\times O}$ (O is the number of observations, equal to the length of the time-series data of EMG). The result is not an exact factorization, but a lower rank approximation to the original matrix. NMF seeks to minimize the cost function expressed as

$$\delta = \frac{1}{2}||f(\mathbf{Z}) - \mathbf{W} \times \mathbf{H}||_F^2 \tag{4}$$

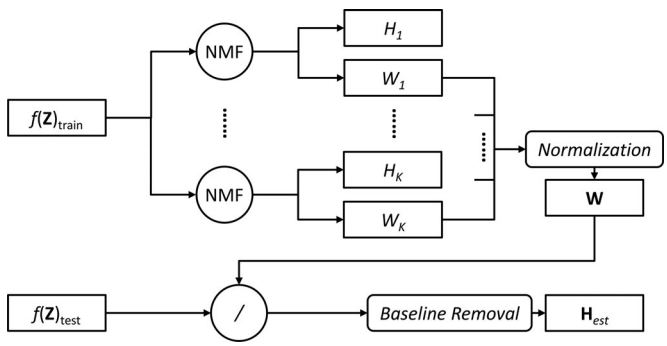

Fig. 1. From the training data, $W_i\,(i = 1, 2, \ldots, K)$ is DOF-wise calculated by individual NMF algorithms, and thus $\mathbf{W}$ is obtained. For new testing data $f(\mathbf{Z})_{\text{test}}$, the estimated synergy activation is $\mathbf{H}_{\text{est}} = \mathbf{W}^{-1} f(\mathbf{Z})_{\text{test}}$.

with the multiplicative updating process

$$\mathbf{H}_{n+1} = \mathbf{H}_n * \frac{\mathbf{W}_n^T \times f(\mathbf{Z})}{\mathbf{W}_n^T \times \mathbf{W}_n \times \mathbf{H}_n + \epsilon}$$
$$\mathbf{W}_{n+1} = \mathbf{W}_n * \frac{f(\mathbf{Z}) \times \mathbf{H}_{n+1}^T}{\mathbf{W}_n \times \mathbf{H}_{n+1} \times \mathbf{H}_{n+1}^T + \epsilon} \tag{5}$$

where in (4) Frobenius norm $||\mathbf{A}||_F = \sqrt{\sum_i \sum_j |a_{ij}|^2}$, and in (5) ϵ is a very small number [22].

The NMF algorithm is naturally unsupervised. However, a complete unsupervised algorithm is not powerful enough to solve the problem. Thus, Jiang *et al.* [20] modified it to create a semisupervised version by calculating W_i "DOF-wisely," which means at any time only one W_i is calculated, but not the whole $\mathbf{W}$. Choi and Kim [21] chose to do this in a supervised way by fixing the initial $\mathbf{H}$ and not updating it during the NMF iterations. In our experiment, we found that the calculation results of these two methods hardly differ. We choose the DOF-wise method for which it is unnecessary to label the training data.

The detailed usage of the NMF algorithm in our application is illustrated in Fig. 1. For the output $\mathbf{H}_{\text{est}}$, it is preferable that the corresponding H value should be equal to 1 if a synergy is fully activated and equal to 0 if a synergy is inactivated. This can be achieved by using normalization and baseline removal. Although, for $\mathbf{H}_{\text{est}}$, the full activation values and the baseline values of synergies are unknown, they can be approximately estimated from $\mathbf{H}_{\text{train}}$ which is equal to $\mathbf{W}^{-1} f(\mathbf{Z})_{\text{train}}$. In the normalization step, $\mathbf{W}$ is scaled so that each W_i is multiplied by $\overline{H_i(t_{\text{act}})}$ (the average value over the "fully activated period" of the ith synergy of $\mathbf{H}_{\text{train}}$). This step ensures that $H_i(t)$ (from $\mathbf{H}_{\text{train}}$) is approximately equal to 1 if the ith synergy is fully activated at time t. As a result, $\mathbf{H}_{\text{est}}$ is also normalized. After normalization, the baseline removal step first records the baseline of each synergy which is defined as $\overline{H_i(t_{\text{rest}})}$ (the average value over the "rest period" of the ith synergy of $\mathbf{H}_{\text{train}}$). Since $\mathbf{H}_{\text{est}}$ is expected to have similar baseline values with $\mathbf{H}_{\text{train}}$, subtracting $\overline{H_i(t_{\text{rest}})}$ from $\mathbf{H}_{\text{est}}$ can remove the baselines.

TABLE I
FEATURES

Feature	Description
MAV	Mean absolute value
MSV	Mean square value, equivalent to variance
RMS	Root mean square, equivalent to standard variance
WL	Waveform length, calculated by $\sum_{k=1}^{N-1} \lvert x_{k+1} - x_k \rvert$
ZC	Zero crossings, the number of times the waveform crosses zero
SS	Slope sign changes, the number of times the slope of the waveform changes its sign
Log	Log-detector, calculated by $\exp(\frac{1}{N}\sum_{k=1}^{N} \log\lvert x_k \rvert)$

B. Feature Selection

Different EMG features have been taken as input to the NMF algorithm, such as root mean square (RMS) [19], mean square value (MSV) [20], and mean absolute value (MAV) [21]. Among these studies, only Jiang *et al.* [20] states that MSV could be considered linearly correlated with muscle activation level $\mathbf{V}$. For (3) to hold, it is important to find proper features which have an approximately linear correlation with $\mathbf{V}$. On the other hand, if too many features are adopted, the decomposed matrices of NMF may also deteriorate because irrelevant input can convey large influence to the output. Jiang's research also mentioned that fewer EMG channels yield better performance when the muscle crosstalk is dominant [20]. This is why a feature selection process is needed to carefully choose the input of the NMF algorithm.

The NMF algorithm requires nonnegative input. Based on this prerequisite, candidates of time-domain features are chosen as in Table I. Most of them are frequently suggested features in pattern recognition studies [2], [23], [24]. Each feature has eight dimensions, corresponding to eight EMG channels. On each channel, all of the features are extracted by using a sliding window with an overlap. The lengths of the window and the overlap are mentioned in Table II. In training phases, each feature is divided by its maximum value so that it is normalized to a range of $[0, 1]$. In test phases, the same values are used to approximately normalize the features.

C. Control Scheme

As illustrated in Fig. 1, after the NMF algorithm calculates the decomposed matrices $\mathbf{W}$ and $\mathbf{H}$, for any new coming data $f(\mathbf{Z})_{\text{test}}$, the estimated synergy activation is $\mathbf{H}_{\text{est}} = \mathbf{W}^{-1} f(\mathbf{Z})_{\text{test}}$. This output $\mathbf{H}_{\text{est}}$ contains information on the time-varying activations of synergies, but with too many disturbing fluctuations. A control scheme is necessary to smooth the $\mathbf{H}_{\text{est}}$ and to qualify it as the direct control signal of the prosthesis. Until now, people have not paid much attention to this problem in laboratory studies, but it is inevitable for an integrated practical system. The traditional idea for smoothing is applying a low-pass filter, but according to our experiments, a low-pass filter is not enough to suppress the fluctuations and help the subject to maintain stable control signals. To prove this, a control scheme for the prosthetic application is customized, and a comparison with a low-pass filter is made in this study.

TABLE II
EXPERIMENT SETTINGS

Set	Device	Electrodes	Sample Rate	Filter	Window Length	Overlap	Time Resolution
#1	NI USB	8-channel bipolar	1000 Hz	30–300 Hz bandpass, 60-Hz notch	200	175	0.025 s
#2	g.USBamp	8-channel bipolar	600 Hz	5–250 Hz bandpass, 60-Hz notch (built-in)	100	85	0.025 s

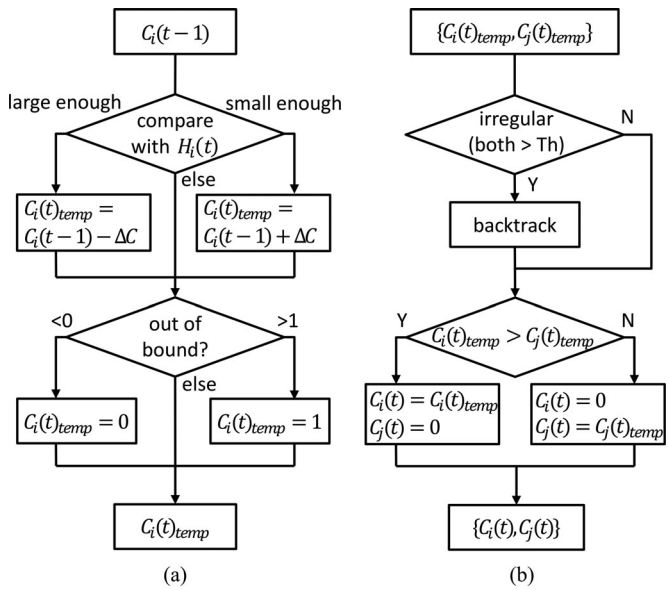

Fig. 2. Block diagram of the control scheme. (a) Step 1 ($i = 1, 2, 3, 4$) and (b) Step 2 ($\{i, j\} = \{1, 2\}, \{3, 4\}$).

Considering the characteristics of the prosthetic application, the following features of the control signal $\mathbf{C}$ are desired.

1) The control signal $\mathbf{C} = [C_1, C_2, C_3, C_4]^T$ where C_1, C_2, C_3, and C_4 are control signals for open, close, pronate, and supinate motions, respectively.
2) The range of $\mathbf{C}$ should be $C_i \in [0, 1]$, where 0 stands for an inactivated synergy (movement) and 1 for a fully activated one.
3) $\mathbf{C}$ should respond to an apparent increase or decrease in $\mathbf{H}_{\text{est}}$ with little delay.
4) $\mathbf{C}$ should be stable, reluctant to change its value if the user is keeping his hand still.
5) Open and close, pronate and supinate cannot be simultaneously activated, which means at the same time, only one of C_1 and C_2, and one of C_3 and C_4 are valid.

According to the rules above, the control scheme algorithm is designed as shown in Fig. 2. The algorithm is composed of two steps. In step 1, first, the conditions for increasing and decreasing the output C_i are defined. Because in this application, a stable output is highly preferred, the change of C_i should be carefully limited. According to the algorithm, only if the value of $H_i(t)$ is larger than $C_i(t-1)$ by a fixed number b_{up}, $C_i(t)$ is allowed to increase. Similarly, only if $H_i(t)$ is smaller than $C_i(t-1)$ by b_{down} (or $H_i(t)$ lower than a predefined baseline, 0.1 in our study), $C_i(t)$ is allowed to decrease. Both the increasing and decreasing of $C_i(t)$ use the same step variable ΔC to constrain the change rate. After that, the lower bound and the upper bound of C_i are applied. Through this process, the values of $C_i(t)(i = 1, 2, 3, 4)$ are temporarily determined.

Step 2 is to check the validity of the synergy pairs, which are open–close ($\{C_1(t), C_2(t)\}$) and pronate–supinate ($\{C_3(t), C_4(t)\}$). First, some irregular cases where $C_i(t)$ and $C_j(t)$ ($\{i, j\} = \{1, 2\}$ or $\{3, 4\}$) are both very large have been observed in experiments. This usually happens when a subject maintains a movement with too much strength, which creates fluctuations on the other synergy. In that case, for example, the current output "100% open" can be suddenly turned to "100% close" (i.e., $C_2(t)$ becomes larger than $C_1(t)$) if the subject opens the hand too hard. To avoid this, in step 2, $C_i(t)$ and $C_j(t)$ are not allowed to go beyond a threshold simultaneously. If this happens, $C_i(t)$ and $C_j(t)$ will be backtracked to their previous states which are $C_i(t-1)$ and $C_j(t-1)$. Finally, because $C_i(t)$ and $C_j(t)$ are mutually exclusive, only the larger one is preserved, and the smaller one is assigned zero.

The parameter values mentioned above are $b_{\text{up}} = 0.4$, $b_{\text{down}} = 0.4$, $\Delta C = 0.05$, and $\text{Th} = 0.7$. These values were determined by trial-and-error.

D. Experiment

1) Apparatus: Our experiments were conducted at two different institutions (The Johns Hopkins University, USA, and Kyoto University, Japan). The two institutions did not share the same experimental devices; therefore, two different sets of devices were used to measure EMG signals. One was NI USB-6009, and the other was g.USBamp with g.GAMMAbox. The details are listed in Table II. Both settings worked properly to measure EMG signals and we found no obvious difference in their performance.

Eight pairs of bipolar electrodes were placed in uniformly spaced circles around the forearm. The electrode positions were not specifically fixed for easy attaching/detaching consideration. Another electrode connected to ground was placed on the elbow near the bone to serve as the reference for the amplifier.

All experiments were performed on a personal computer platform. The source code was written in MATLAB 2011b leveraging the MATLAB Simulink environment. MSMS [25] supported the visualization of a virtual prosthesis on the computer screen. The virtual prosthesis has a large number of DOFs. It receives input commands through UDP ports and is animated in real time. In training mode, it can provide visual cues to instruct the subject to make movements, and in testing mode, it can work like a real prosthesis giving the subject visual feedback. The experimental setup is shown in Fig. 3, and a more detailed view of the virtual prosthesis is shown in Fig. 4.

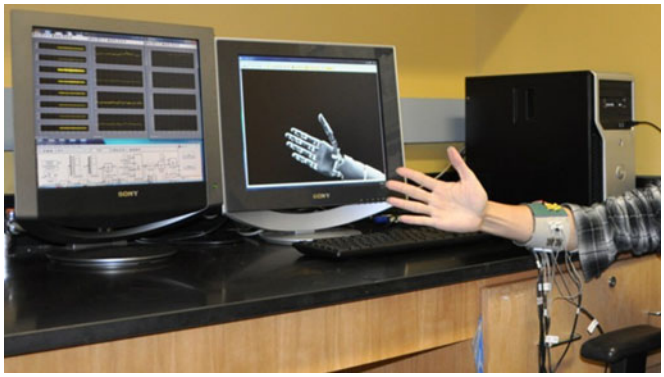

Fig. 3. Experimental setup (corresponding to #1 in Table II).

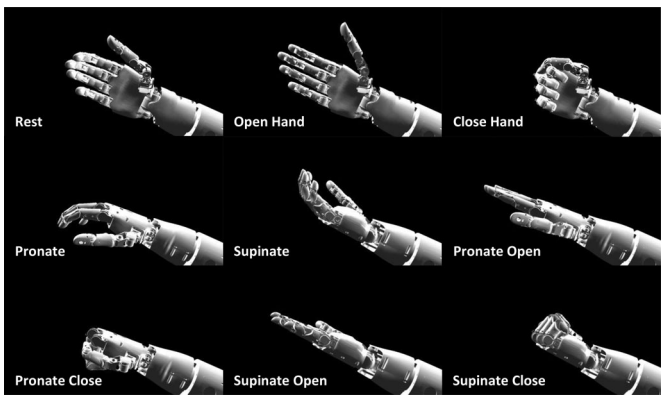

Fig. 4. Virtual prosthesis generated by the musculoskeletal modeling software (MSMS) [25]. In the offline experiment, the static pictures of these nine movements are displayed as cues. In the online experiment, coherent animation of the virtual prosthesis is used as the visual feedback.

2) Participants: Two kinds of experiments were conducted. One is offline, for data analysis, feature selection, and NMF parameter adjustment. The other is online, for evaluating the real-time performance and robustness with the NMF algorithm and the proposed control scheme.

Four healthy subjects participated in the offline experiment. One of them and another nine healthy subjects (total of 10) participated in the online experiment. In both experiments, subjects sat in a chair in front of a computer screen and rested their elbow (left or right, of their choice) on a flat surface.

3) Procedures (Offline Experiment): In the offline experiment, the subject performs nine movements: rest, open, close, pronate, supinate, open and pronate, close and pronate, open and supinate, and close and supinate in a random sequence.

Performing one movement is treated as one trial. It lasts for 9 s and consists of four steps: preparation, movement, hold, and return. Visual cues, as shown in Fig. 4, are presented at the beginning of each trial, and the hint texts "ready," "begin," "hold," and "return" are also displayed together with cues. During the first 2 s, the subject is notified what movements he is supposed to make. During the next 2 s, the subject gradually performs the movement until it reaches the target position

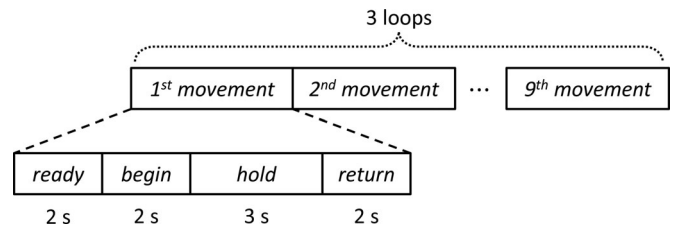

Fig. 5. Flowchart of the offline experiment. The sequence of movements is random.

with a certain strength. After that, he holds the position for 3 s. In the final 2 s, he returns his hand to the initial rest state. Nine trials, which consist of nine randomly ordered movements, compose a subsession. The subject needs to complete three consecutive subsessions as a whole session. Therefore, the full length of one experiment session is $9 \times 9 \times 3 = 243$ s, as shown in Fig. 5.

Each subject completed at least three sessions, using medium strength, low strength, and high strength, respectively. This range was used to determine the best strength for training.

4) Procedures (Online Experiment): The online experiment includes one training session and two test sessions.

The first step is a training session, which is a simple version of the offline experiment. The subject only needs to perform five basic movements in a row: rest, open, close, pronate, and supinate. One trial containing one movement still lasts for 9 s, therefore, it takes 45 s for the whole session. In these trials, the subject makes a movement at the very beginning and returns to a rest state at the last second without cues. The first trial "rest" is only used to record the baseline, and the other four trials are used to calculate the synergy matrix $\mathbf{W}$. $\mathbf{W}$ has four columns ($W_i, i = 1, \ldots, 4$), which represent four synergies. Each W_i is calculated from one trial of training data by using the NMF algorithm.

After the subject completes the training session, he practices to become familiar with the real-time prosthesis control, and then does the online tests.

The tests are done to verify how precisely the subject can control the prosthesis with the proposed algorithm and the control scheme, and whether the subject can maintain the control signal in a stable manner. For this purpose, we instruct the subject to look at a scope window, which dynamically displays the value of the four-channel control signal $\mathbf{C}$ on the screen. This way, the subject can be aware of the "exact value" (in a range of $[0, 1]$) reflected by his movement.

In test 1, the subject is to perform four basic movements (open/close/pronate/supinate) at different strength levels. The sequence of movements is fixed as "open–close–pronate–supinate." For each movement, the subject completes three trials (9 s each) from low strength to high strength. As the y-axes of the scope window are divided into five intervals from 0 to 1 by the division of 0.2, the subject is asked to try to make the control signal C_i fall into the second interval ($[0.2, 0.4]$), the fourth interval ($[0.6, 0.8]$), and reach the maximum ($[1, 1]$), respectively. The signal should be maintained as stable as possible until the end

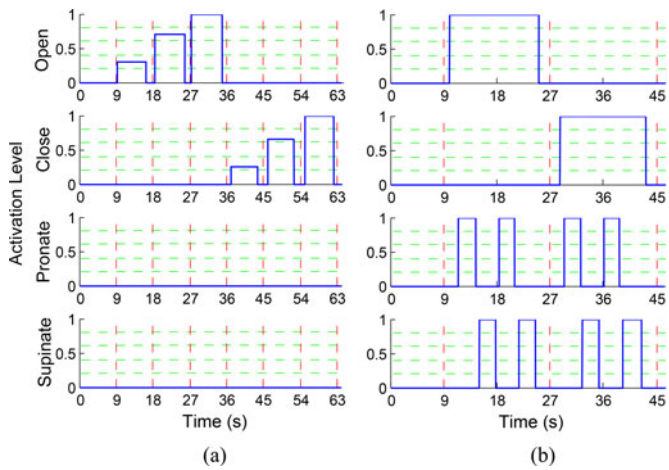

Fig. 6. Ideal demonstrations of online (a) test 1 and (b) test 2. Blue lines are the ideal output of control signals C_1, C_2, C_3, and C_4. Only the first halves of the two tests are shown.

of seventh second. The first half part of test 1 is demonstrated in Fig. 6(a). The full length of test 1 is $9 + 4 \times 3 \times 9 = 117$ s (an additional trial at the beginning is for preparation).

In test 2, we verify the subject's ability to control the four varieties of hand/wrist combined movements: pronate open, pronate close, supinate open, and supinate close. Since there is strong mutual interference between the synergies of hand movement and the synergies of wrist movement, it is very difficult for subjects to precisely control the combined movements. Therefore, we do not require the subject to make movements by various strength levels, but only by the maximal strength.

Test 2 has four trials. In each trial, the subject first starts a primary movement, keeps it at maximum, and then combines it with two secondary movements twice. In the first trial, the subject first activates "open" to maximal level, then activates "pronate–supinate," and then again, without releasing the "open." Each combined movement should be kept at maximum for 2 s at least. The timing of making each movement is controlled by the subject himself as long as he completes the series of movements in one trial of 18 s. The sequence of the trials is still "open (with combinations, the same below)," "close," "pronate," and "supinate." The first half of test 2 is demonstrated in Fig. 6(b). The full length of test 2 is $9 + 4 \times 18 = 81$ s (including the first trial for preparation).

5) Measures: The overall performance of both the offline and online experiments is evaluated by obtaining the R^2 value used in previous studies [12], [20], [21]. It expresses the relative distance between the estimated value and the target value

$$R^2 = 1 - \frac{\sum_{k=1}^{K} \sum_{t}^{T} (\widehat{H_k(t)} - H_k(t))^2}{\sum_{k=1}^{K} \sum_{t}^{T} (H_k(t) - \overline{H_k(t)})^2} \tag{6}$$

where $H_k(t)$ is the target value for kth synergy at time t, whose temporal average is $\overline{H_k(t)}$. $\widehat{H_k(t)}$ is the corresponding estimation, equal to the (k, t)th element of $\mathbf{H}_{\text{est}}$ (offline), or $C_k(t)$ (online). The numerator is the total mean-squared error of the estimates, and the denominator is the total variance. R^2 value is a global measurement of estimation performance. Larger values mean the estimation is closer to the target with an upper bound of unity.

In the offline experiment, the data are simply labeled such that $H_k(t)$ has only two values: 0 for inactive and 1 for active. In the online experiment, except for the target value 1 representing full activation, we have two target intervals $[0.2, 0.4]$ and $[0.6, 0.8]$. The most common values appearing in those intervals are taken as the corresponding label value. For example, for one trial, if the target was $[0.2, 0.4]$, and the subject held his movement mostly at 0.35, then the label value should also be 0.35; if the subject held his movement mostly at 0.5, the label should be equal to 0.4 because it cannot be out of range.

T in (6) represents the time intervals of interest. The value is determined by each experiment setting. For the offline experiment, it is the 5–7th seconds in each trial when the cue is "hold." For online test 1, T is expanded a little longer to 4–7th seconds in each trial, where the 1–3rd seconds are for the subject to adjust his movement. For online test 2, T should be a 2-s interval when the combined movements are fully activated. In this test, the starting point of each movement is uncertain and is manually labeled over the data.

6) Data Analysis: Four subjects completed 14 offline sessions, which include five sessions at medium strength, four sessions at low strength, and five sessions at high strength. Fig. 7 shows an example of one complete offline session. Note that in the offline experiment, the subject did not receive visual feedback, the control scheme was not used, and the baseline removal was also not applied. In the figure, there are a lot of fluctuations, as well as nonzero baselines. In that case, the R^2 value can hardly achieve a high level. But as our main target is to compare the different features, these negative effects can help us find the robust ones. Despite these limitations, the synergies of "close" and "pronate" have ideal shapes in the figure, and Fig. 7(a) also shows that the nine movements are well represented by average synergy activations.

After analyzing the data of the offline experiment, the performance of different features is shown as in Fig. 8. Through comparing the maximal, minimal, and mean R^2 values of different features, it can be concluded that the best three features are MAV, RMS, and WL. Log and MSV are nearly as good. ZC and SS are frequently used in classification methods, but are obviously not oriented to the NMF algorithm.

As per the results of the offline experiment, the three best performing features: MAV, RMS, and WL, are adopted in the online experiment. The total dimension of the feature set is 24 (3×8 channel). Because features have different ranges, each one is normalized by a scaling value equal to the maximum of that feature in the training data.

III. Results

Fig. 9 shows an example of the training result. In the figure, red lines are synergy activation levels which are calculated by applying the NMF algorithm to the input EMG data, and blue lines are the output of the proposed control scheme. Whenever a training session is completed, this figure will be displayed for

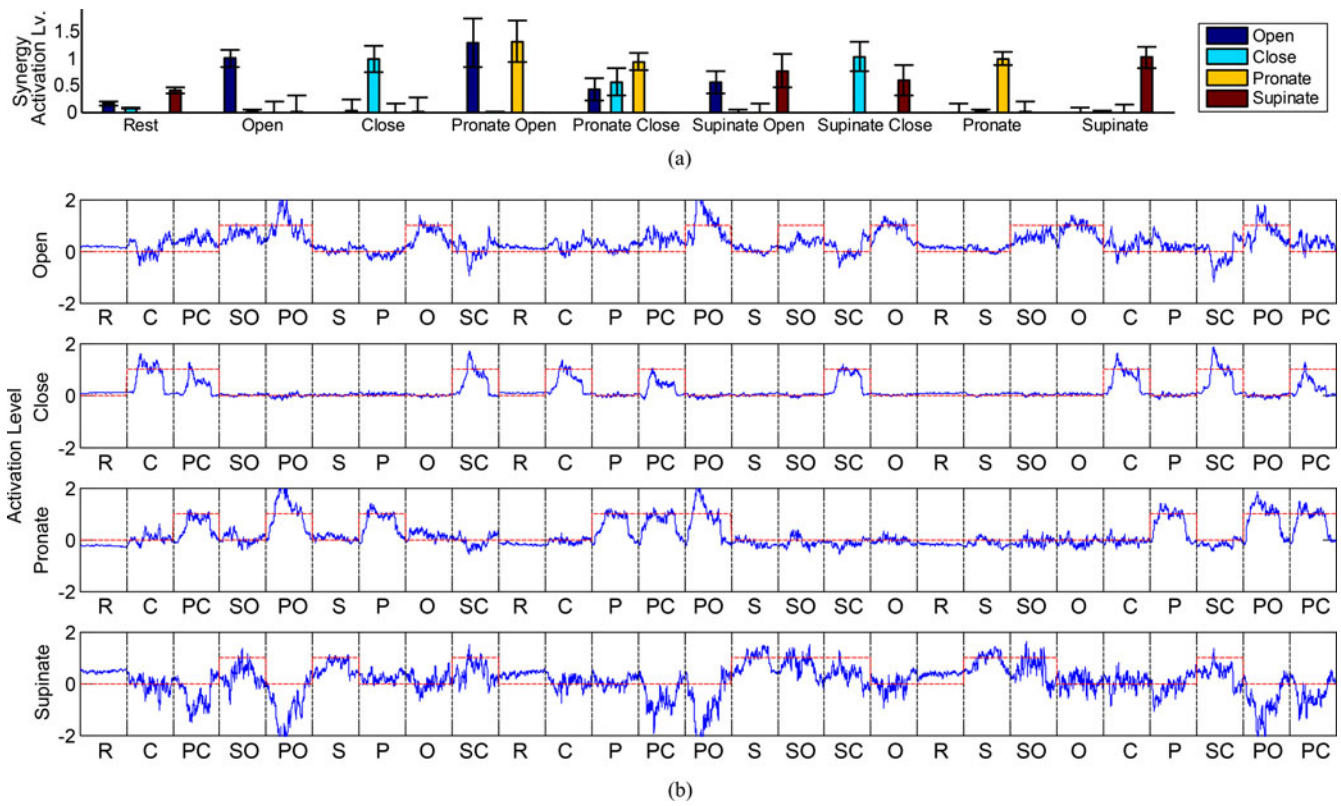

Fig. 7. One session of the offline experiment. (a) Shows the average synergy activation levels for nine different movements in the experiment and their standard variations. In (b), the blue lines are the time-varying synergy activations ($\widehat{H_k(t)}$ in (6)) in trials (the first 2 s for preparation in each trial is omitted). The labels of x-axes represent different movements (R for rest, O for open, C for close, P for pronate, and S for supinate). The red dash lines are the target activation levels ($H_k(t)$ in (6)).

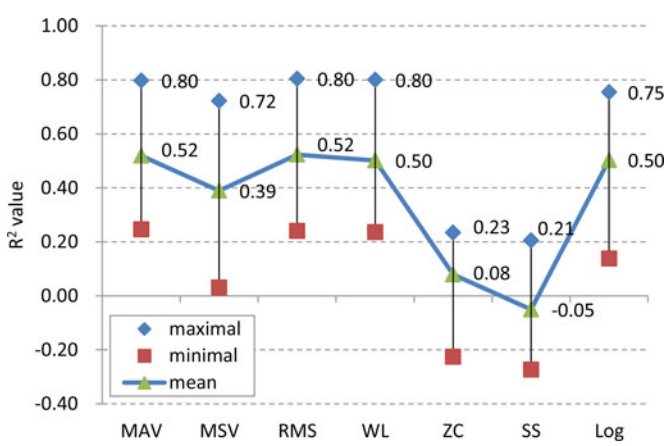

Fig. 8. Maximal (blue diamonds), minimal (red squares), and mean (green triangles) R^2 value of 14 offline experiment sessions for seven features.

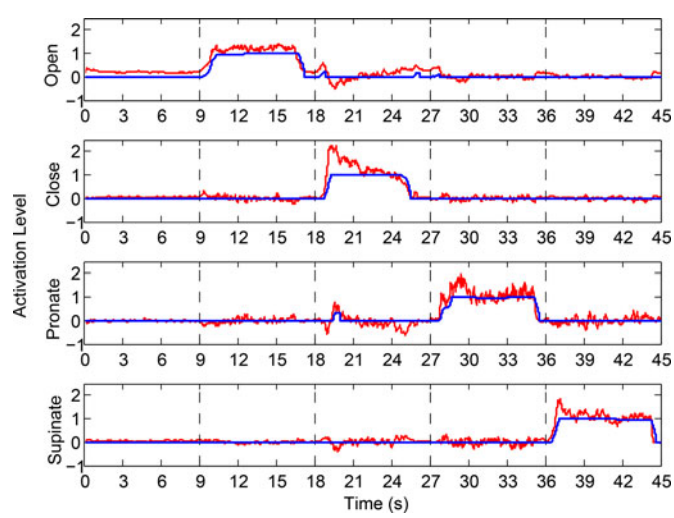

Fig. 9. Example of training session data from the online experiment. Red lines are activation level of four synergies, and blue lines are corresponding control signals C_1, C_2, C_3, and C_4.

evaluating the quality of training. Relatively smooth red lines (see Fig. 9) indicate high training quality.

Fig. 10(a) shows an example of online test 1, where the subject completed the task well. There is little redundant control or noise during the whole test. With the exception of the first pronate trial, which is slightly overactivated, other movements fall in the desired intervals, and they are also stably maintained in most trials. This example shows NMF is evidently capable of fluidly controlling individual DOF.

Fig. 10(b) shows an example of online test 2. The subject was also able to finish the task adequately. Generally, multiple

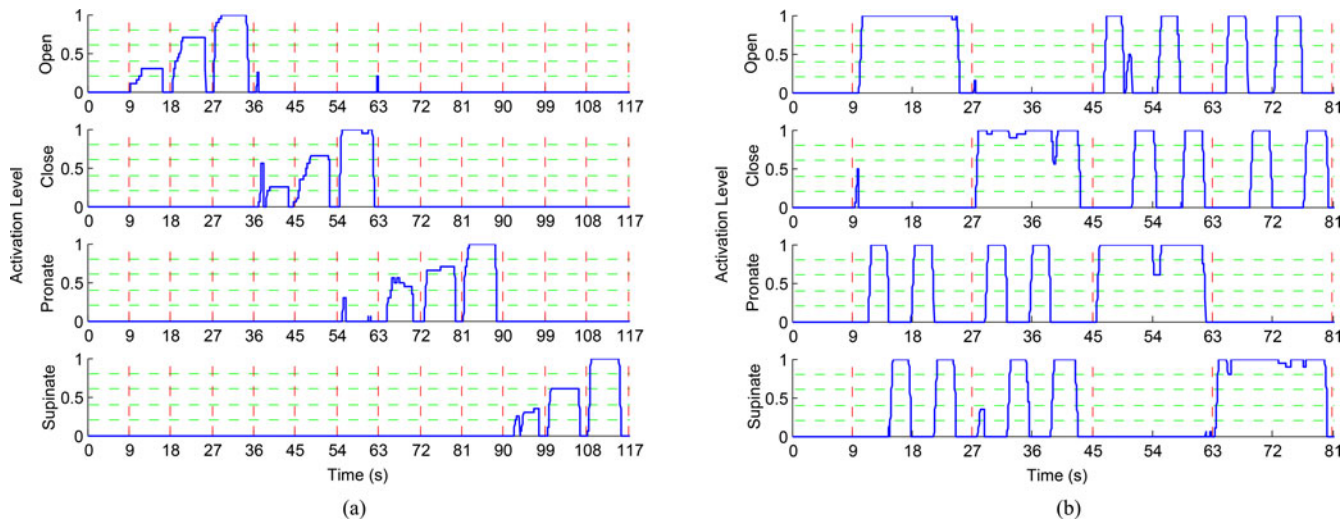

Fig. 10. Examples of online (a) test 1 and (b) test 2. Blue lines are the system output (control signals C_1, C_2, C_3, and C_4) which the subject can see from the screen during the experiment.

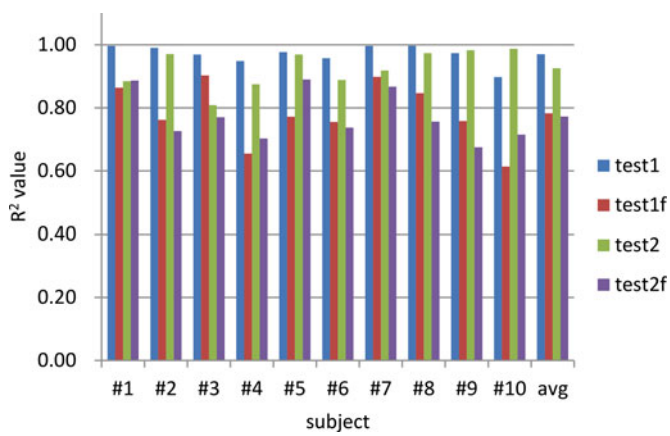

Fig. 11. R^2 value calculated from all the ten subjects' online experimental data (test 1 and test 2), as well as from the same data processed by low-pass filter, but not our control scheme (test 1f and test 2f). The average R^2 values of test 1 and test 2 are 0.97 and 0.93, respectively.

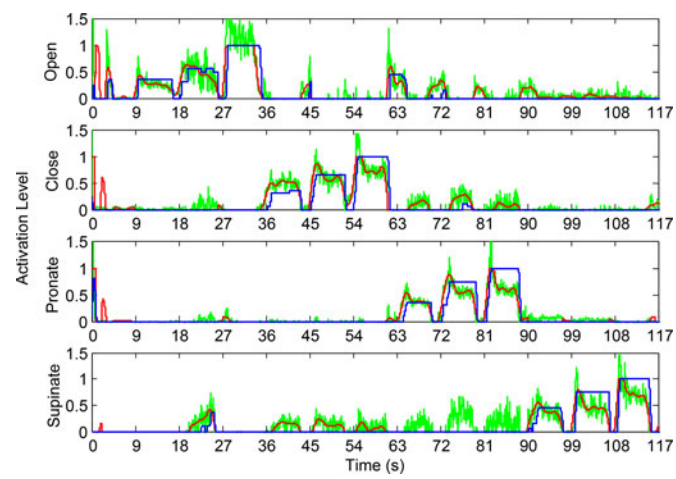

Fig. 12. One example of the experiment data of test 1 for comparison. Green lines are estimated synergy activations ($\mathbf{H}_{\rm est}$). Red lines are $\mathbf{H}_{\rm est}$ processed by the low-pass filter (with a output range [0,1]). Blue lines are $\mathbf{H}_{\rm est}$ processed by the control scheme (control signals C_1, C_2, C_3, and C_4).

DOFs are more difficult to control than an individual DOF is, because wrist and hand movements are likely to interfere with each other. To simultaneously control two DOFs precisely is still too challenging to complete; thus, we only require the subject to achieve maximal activations in this test.

Fig. 11 shows the evaluation of the online experiment. The average R^2 values of test 1 and test 2 among 10 subjects achieve 0.97 and 0.93, respectively, while using a low-pass filter the R^2 values are 0.78 and 0.77. The result indicates that the proposed control scheme always has a satisfactory result and outperforms the low-pass filter. Note that online filters have significant time delays; therefore, in this comparison, an offline zero-phase FIR filter (0.5-Hz cutoff) is applied.

Fig. 12 shows a detailed example of the comparison between the original data, the output of the filter, and the output of the control scheme for test 1. The figure shows that the filter works well for eliminating the high-frequency noise, but is unsuitable for the task of "making movements and maintaining them." From each trial in the figure, although the subject was told to maintain the movement, $\mathbf{H}_{\rm est}$ has a trend to spontaneously decrease after it reaches a maximum, while the proposed control scheme can effectively prevent the signal from an unwanted drop most of the time. Of course, the drop does not occur every time (see 36–45 s of "close"), and using the online filter, the subject may give a better response to compensate for the drop. However, it is still difficult to suppress fluctuations without using the control scheme.

Besides maintaining the movement and reducing the fluctuations, the proposed algorithm can also suppress the wrongly activated synergy to some extent. See the fourth plot in Fig. 12, as an example. According to our experiments, the "supinate" synergy should be only active in the interval of 90–117 s. However, green lines indicate that the "supinate" synergy has

been wrongly activated in the intervals of 18–27 s and 36–90 s, which are seven trials in total. Among them, the last three trials (63–90 s) were not truly affected because the "pronate" synergy was dominant during that period. For the other trials, the output of the low-pass filter (red lines) just reproduced the shape of the affected data, whereas the control scheme kept a nearly desirable zero output.

IV. Discussion

This study proposed an NMF method coupled with a smoothing postprocessing algorithm to accurately control a virtual prosthesis by the forearm surface EMG. Four hand/wrist movements: open, close, pronate, and supinate have been decoded and the proportional control of multiple DOFs have been realized. We have carried out an offline experiment to find the optimal EMG features for NMF decomposition. Two online tests: 1) proportional control on single DOF, and 2) simultaneous control of multiple DOFs, have been conducted to evaluate the proposed method.

A. Offline Experiment

The results of offline experiments indicate that time-domain features (especially those with first-order property) are suitable to NMF, but frequency-domain ones may not be. Through the experiments, we also found that MAV and RMS features only perform well when the signal has a zero mean, which means it is better to either remove the dc component beforehand, or to subtract the mean values when calculating the features. On the other hand, WL are robust for signals with either zero mean or nonzero mean, since its definition has no relation with mean values.

Aside from the performance of different features, the experiments also reveal some other information. First, it is difficult to judge which is better, high strength or normal strength, but it is certain that using low strength always leads to the worst result. Second, the initial condition of the NMF algorithm can vary, for example, whether $f(\mathbf{Z})$, the input of NMF, should be trimmed so that only fully activated intervals are preserved; whether the $\mathbf{H}$ should be given an initial value, or even be kept unchanged during the NMF updating process. Tests on these various initial conditions have been made and no significant differences were found. The initial condition is most likely to affect the scaling of the output, but not its trend over time.

B. Online Experiment

Judging from the performance of the subjects, nearly 100% completed test 1 with high accuracy, although some needed more practice than others. There are two problems identified by the online experiments. The first is when activating one movement, unrelated movement from different synergies may also be activated by chance. The second is that the activation level is difficult to keep stable at low strength levels. For the first problem, through practice, the subject can gradually learn how to suppress unrelated movements. For the second one, the poor performance is mainly due to noise (fluctuations) in $\mathbf{H}_{\mathrm{est}}$. With the low strength level, the input data fluctuations near the baseline are difficult to correct with the control scheme. Repeating the training process helps solve this problem.

Test 2 is more difficult for most subjects. Although, the individual movements are likely to be completed without problems, activating and maintaining two movements simultaneously are different from performing them individually. In that case, the muscle crosstalk will have considerable influence on the result. When the subject is maintaining one movement and tries to activate another, the first movement becomes uncontrollable and its level drops. In our experiments, all the subjects successfully completed test 2, but the overall practice time was longer than that of test 1.

C. Practicality

For surface EMG measuring, every time the electrodes are detached/attached again, the electrodes' positions change, which requires training again. Moreover, after wearing the prosthesis for a period of time, the electrodes may be slightly displaced, and the states of the skin and muscle may also be changed (due to fatigue, temperature, etc.). As a result, the original training data are likely to become unsuitable over time. Until now, there has been no reliable solution to prevent the original training data from degrading. One alternative is to shorten the training process so that the system can recover. In our experiment, the training time is 54 s in total, 9 s for each movement (including rest). Shorter training time is also possible, but has not been verified. Moreover, the direct cause of performance drop is baseline drift. Having a countermeasure against baseline drift may help avoid retraining, and further raise the efficiency.

It is not clear how many DOFs can be supported by this kind of prosthesis control method. Currently, the 4-movement control paradigm (open/close/pronate/supinate) tested in this study has acceptable performance. However, if the number of synergies increase with DOFs, subjects would be more likely to activate unrelated synergies when performing any movement. A better solution may be to find a set of synergies whose number is less than the number of DOFs, where such synergies form basis vectors for a wide variety of movements. In [19], Ajiboye and Weir used an NMF-based method to build a predictive framework for 33 hand postures with only eight synergies. Their classification method, however, does not involve proportional control. When synergies and movements do not have one-to-one correspondence, the usage of the activation levels of multiple synergies to represent degrees of a single movement is an issue to be addressed.

In this study, considering the easy attaching/detaching property of the prosthesis, the circularly arranged electrodes are fixed at the proper position on the forearm. In other studies, the electrodes were often placed on specified muscle sites. Both settings have their advantages and disadvantages. Setting the electrodes on the correct sites require a great deal of preparation, especially for amputees; therefore, the specified electrodes placement is more applicable to implantable myoelectric sensors than to surface EMG. However, we found that the specified electrodes placement does improve the quality of signals,

because it lowers the muscle crosstalk. If conditions permit or good performance is highly desired, the specified electrode placement can be taken into consideration.

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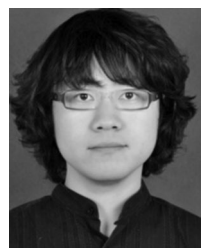

Jiaxin Ma received the B.S. and M.S. degrees in computer science from Shanghai Jiao Tong University, Shanghai, China, in 2008 and 2011, respectively, and the M.S. degree in engineering from Waseda University, Tokyo, Japan, in 2009. He is currently working toward the Ph.D. degree with the Department of Mechanical Engineering and Science, Kyoto University, Kyoto, Japan.

He was a Visiting Scholar with the Department of Biomedical Engineering, Johns Hopkins University, Baltimore, MD, USA in 2012. His current research interests include brain-computer interface, electroencephalogram, electrooculogram, and electromyogram signal processing and their applications.

Nithish V. Thakor (S'78–M'81–SM'89–F'97) received the undergraduate degree from Indian Institute of Technology Bombay, Mumbai, India, in 1974, he worked as an engineer with Philips India for two years, and he then received the Ph.D. degree from the University of Wisconsin, Madison, WI, USA, in 1981.

From 1981 to 1983, he was a Member of the faculty of Northwestern University, and since then he has been with the Johns Hopkins School of Medicine. He was a Provost Professor, National University of Singapore, and is currently the Director of Singapore Institute for Neurotechnology, Singapore, an institute dedicated to neurotechnology research and development. He is also a Professor of Biomedical Engineering, Electrical and Computer Engineering, and Neurology at Johns Hopkins and directs the Laboratory for Neuroengineering. His technical expertise is in the areas of neural diagnostic instrumentation, neural microsystem, neural signal processing, optical imaging of the nervous system, rehabilitation, neural control of prosthesis and brain machine interface.

Fumitoshi Matsuno (M'94) received the Ph.D. (Dr. Eng.) degree from Osaka University, Osaka, Japan, in 1986.

Since 2009, he has been a Professor with the Department of Mechanical Engineering and Science, Kyoto University, Kyoto, Japan. He is currently the Vice-President of NPO International Rescue System Institute. His current research interests include robotics, swarm intelligence, control of distributed parameter system and nonlinear system, and rescue support system in disaster.